Lab 05: Simple Regression.

Question 1. Model Specification

For the questions below, specify a Model A/Model C comparison that would allow you to address the hypothesis in question.

- In a randomized controlled trial, we are looking at the effects of a resistance training intervention on osteoarthritis in older women. We have change scores in bone mineral density from pre- to post-intervention (a variable called 'CHANGE') and we want to know if there was an effect of GROUP (a resistance training group compared to an aerobic exercise control group).
    - MODEL A: $CHANGE_i = \beta_0 + \beta_1(Group) + \varepsilon_i$
    - MODEL C: $CHANGE_i = \beta_0 + \varepsilon_i$
- There is a known relationship between AGE and BMI in children. Assume we have a sample of children from the Utah public school system. Controlling for the relationship between BMI and AGE, we want to know if the average child in our sample has a BMI of less than 30.
    - Note that BMI = Body Mass Index = kg_i/m_i^2
    - MODEL A: $BMI_i = \beta_0 + \beta_1(Age_i - \overline{Age}) + \varepsilon_i$
    - MODEL C: $BMI_i = 30 + \beta_1(Age_i - \overline{Age}) + \varepsilon_i$
- Broadly speaking, there is a negative relationship between tree-height (HEIGHT) and elevation (ALTITUDE). What models would we compare to see if this relationship was statistically different from zero?
    - MODEL A: $Height_i = \beta_0 + \beta_1(Altitude_i) + \varepsilon_i$
    - MODEL C: $Height_i = \beta_0 + \varepsilon_i$
- Researchers have been studying the effects of land elevation on tree growth for many years. Let's assume that a researcher has a very specific hypothesis that this slope is -1.75. Using the same variables as above, what two models would we need to compare to see if this hypothesis is could be rejected?
    - MODEL A: $Height_i = \beta_0 + \beta_1(Altitude_i) + \varepsilon_i$
    - MODEL C: $Height_i = \beta_0 - 1.75(Altitude_i) + \varepsilon_i$

Question 2. Calculating Slopes and Intercepts

In the same way we can use the sum of squared errors as an estimate to get a mean (i.e., the single value that produces the smallest sum of squared errors), we can also use the sum of squared errors to estimate a slope and an intercept (i.e., the two values that produce the smallest sum of squared errors in a linear model).

Eq 1. $slope = b_1 = \frac{\sum(x_i-\bar{x})(y_i-\bar{y})}{\sum(x_i-\bar{x})^2}$

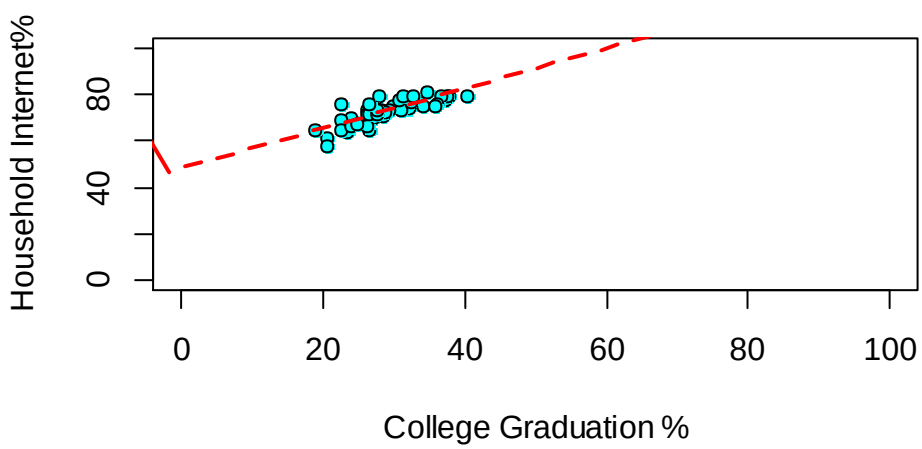
Eq 2. $intercept = b_0 = \bar{y} - b_1(\bar{x})$

```
> summary(mod1)
Call:
lm(formula = Internet ~ College, data = DAT1)

Residuals:
   Min     1Q Median     3Q    Max 
-8.094 -1.722 -0.175  1.470  8.296 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 47.91227    2.69854  17.755  < 2e-16 ***
College      0.86185    0.09213   9.354 2.15e-12 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 3.162 on 48 degrees of freedom
Multiple R-squared:  0.6458,	Adjusted R-squared:  0.6384 
F-statistic:  87.5 on 1 and 48 DF,  p-value: 2.15e-12
```



Using the example from the textbook, we want to estimate the INTERNET access rate in each state as a function of COLLEGE graduation rates. Using R, calculate each of the values below

A. $\bar{y} =$

```
> mean(DAT1$Internet)
[1] 72.806
```

B. $\bar{x} =$

```
> mean(DAT1$College)
[1] 28.884
```

C. $\sum(x_i - \bar{x})(y_i - \bar{y}) =$

```
> DAT1$x_res<-DAT1$College-mean(DAT1$College)
> DAT1$y_res<-DAT1$Internet-mean(DAT1$Internet)
> DAT1$xy_cov<-DAT1$x_res*DAT1$y_res
> sum(DAT1$xy_cov)
[1] 1015.285
```

D. $\sum(x_i - \bar{x})^2 =$

```
> sum((DAT1$College-mean(DAT1$College))^2)
[1] 1178.027
```

E. $b_1 =$

```
> DAT1$x_var<-DAT1$x_res^2
> sum(DAT1$xy_cov)/sum(DAT1$x_var)
[1] 0.8618517
```

F. $b_0 =$

```
> mean(DAT1$Internet)-0.8618517*mean(DAT1$College)
[1] 47.91228
```

G. Provide a written interpretation of this slope and intercept. That is, how should we interpret these two values?

The intercept indicates that for a state with a college graduation rate of 0%, the predicted internet access rate is 47.9% of households. The slope indicates that for every 1-percent increase in college graduate rates, the estimated increase in Internet access is 0.86 percent.

Next, we want to rescale college graduation rates to that they are a proportion rather than a percentage. That is, the scale of COLLEGE will change from 0 to 100, to 0 to 1. Using the lm() function in R, what are the new values of the slope and intercept?

```
> DAT1$College.p<-DAT1$College/100
> mod4<-lm(Internet~College.p, data=DAT1)
> summary(mod4)

Call:
lm(formula = Internet ~ College.p, data = DAT1)

Residuals:
   Min     1Q Median     3Q    Max 
-8.094 -1.722 -0.175  1.470  8.296 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)   47.912      2.699  17.755  < 2e-16 ***
College.p     86.185      9.213   9.354 2.15e-12 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 3.162 on 48 degrees of freedom
Multiple R-squared:  0.6458,	Adjusted R-squared:  0.6384 
F-statistic:  87.5 on 1 and 48 DF,  p-value: 2.15e-12
```

H. $b_1 = 86.185$

I. $b_0 = 47.912$

J. Provide a written interpretation of this slope and intercept. That is, how should we interpret these two values?
When we have rescaled the college graduation rates (0 to 1, rather than 0 to 100), the intercept of 47.9 indicates the predicted internet access rate for a state with a college graduation rate of zero. The slope of 86.2 however, indicates the estimated change in internet access rates for a 100% change in college graduation rates.

Finally, we provide the code to "mean-center" college graduation rates, $x_{ci} = x_i - \bar{x}$. Using the lm() function in R, what are the new values of the slope and intercept?

```
> DAT1$College.c<-DAT1$College-mean(DAT1$College)
> mod2<-lm(Internet~College.c, data=DAT1)
> summary(mod2)

Call:
lm(formula = Internet ~ College.c, data = DAT1)

Residuals:
   Min     1Q Median     3Q    Max 
-8.094 -1.722 -0.175  1.470  8.296 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 72.80600    0.44722 162.799  < 2e-16 ***
College.c    0.86185    0.09213   9.354 2.15e-12 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.162 on 48 degrees of freedom
Multiple R-squared:  0.6458,	Adjusted R-squared:  0.6384 
F-statistic:  87.5 on 1 and 48 DF,  p-value: 2.15e-12
```

K. $b_1 = 0.86$

L. $b_0 = 72.8$

M. Provide a written interpretation of this slope and intercept. That is, how should we interpret these two values?

When we have re-centered the college graduation rates around the mean college graduation rate (28.9), the interpretation of the slope stays the same: 0.86 is the predicted increase in internet access when college graduation rates increase by 1-percentage point. However, the intercept now has a new meaning, as the intercept of 72.8 is the predicted internet access rate for a state with the mean college graduation rate.

Question 3. Statistical Inferences about Parameters

A. Write out the two models we are comparing when we want to test the hypothesis that there is a relationship between INTERNET access rates and COLLEGE graduation rates.

- MODEL A: $Internet_i = \beta_0 + \beta_1(College_i) + \epsilon_i$
- MODEL C: $Internet_i = \beta_0 + 0(College_i) + \epsilon_i$

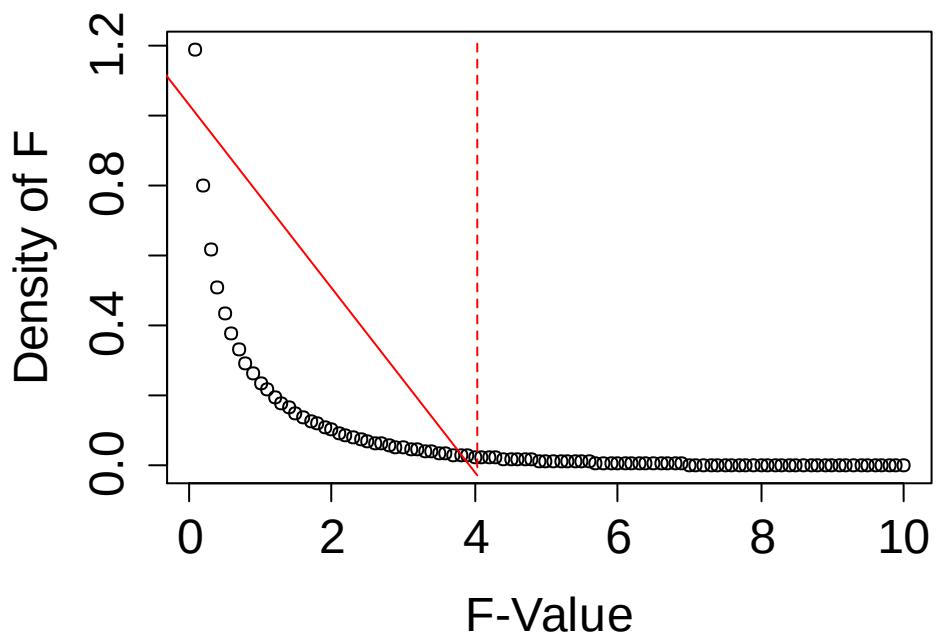
B. What is the proportional reduction in error (PRE), when you compare these two models?

SSE(C) = 1355.028

SSE(A) = 480.0041

$$PRE = \frac{SSE(C) - SSE(A)}{SSE(C)} = 0.6457608 \approx 64.6\%$$

C. Under the null hypothesis, what does the expected F-distribution look like? Insert a plot showing the null F-distribution. Include a vertical line indicating the F-critical value?



D. Using the PRE you calculated above and the appropriate degrees of freedom, what is the **F-observed** value for this model comparison?

```
> # Calculation of F_observed
> # df1 = PA - PC
> df1 <- 2-1
> df1
[1] 1
> # df2 = N - PA
> df2 <- length(DAT1$Internet)-2
> df2
[1] 48
> F_obs<-(PRE/df1)/((1-PRE)/df2)
> F_obs
[1] 87.50168
```

E. Comparing the F-critical value to the F-observed value, would you reject the null hypothesis in this situation?

Yes, the F_obs of 87.5 is greater than the F_critical of 4.042652.

F. Finally, if you wanted to test the null hypothesis for the intercept, rather than the slope, what two models would we need to compare?

a. MODEL A: $Internet_i = \beta_0 + \beta_1(College_i) + \epsilon_i$

b. MODEL C: $Internet_i = 0 + \beta_1(College_i) + \epsilon_i$